

Post-doctoral position in ecology or fisheries science

Personnal statement: passionate about ecology, I am particularly interested in aquatic species and how global changes can impact them (demographics, life history traits, etc.). I have specialised in modelling based on large datasets of various kinds, paying close attention to the biases associated with them. I am looking for a new post-doctoral position



Work experience:

2024 - 2025 (15 months) Post-doctoral position on the passability of a hydroelectric dam by yellow European eels *Anguilla anguilla* at EABX (INRAE, Cestas, France), (Supervision: Hilaire Drouineau):

Using a Bayesian framework, I am developing a **mechanistic capture-recapture** model to better understand the contribution of multiple eel ramps in **optimising** the passage of eels through large hydroelectric facilities. In order to do this, I explicitly model the different **processes**, and in particular the **effect of the environment** on the migration decision and the passage of ramps.

I **manage** the project and **organise** meetings with the various participants: Romain Roy (Electricité de France EDF), Pierre Sagnes (Office français de la biodiversité, OFB), Vanessa Lauronce (Migado)

I am writing a **technical report** for the project applicant, EDF, as well as a **scientific publication**.

I participated and presented at the **FSBI 2025 Conference**.

I also participated in **fieldwork** aboard a boat: I helped **identify, measure, and count** fish.

2021 - 2024 (3 years) PhD on the phenology migration of the Atlantic salmon *Salmo salar* at ECOBIOP (University of Pau and the Adour Region, INRAE, Saint-Pée-sur-Nivelle, France), (Supervision: Mathieu Buoro, Etienne Prévost):

I developed **mechanistic capture recapture** models in **Bayesian** framework to better estimate the number of migrants and to study their **phenology**. I have been careful to disentangle the different **mechanisms** involved, particularly how the **environment** can affect both the biological process and the observation process. I focused on the various **biases** that can occur when monitoring phenology and/or migration, and suggested solutions for my case study of smolt migration in the River Scorff (France) over a 25-year period.

I **organised** several seminars and workshops, including the **international workshop NoWPas 2023** (<https://nowpas.wordpress.com/>) in Norway.

I have attended and presented at a number of **national and international** conferences and seminars.

I got the **FSBI 2023 Conference Award** to participate and present my work.

I **taught** at the University of Pau and the Adour Region and participated to science **communication** event for the general public.

I participated to **fieldwork**, especially electric fishing and marking.

I have attended a number of training courses on a variety of subjects such as **scientific integrity**, best practices for **reproducible research** in digital ecology, **teaching**, **project management**, etc.

2021 (6 months) Internship on the seasonal migration of the European seabass *Dicentrarchus labrax*, at IFREMER (Plouzané, France), the French Research Institute for the Exploitation of the Sea, (Supervision: Mathieu Woillez, Michel Bertignac):

I worked on a **capture recapture** coupled with a **population dynamics** model to estimate the **migration rate** between the Northern stock and the Southern stock of the European seabass. The main objectives was to better understand the population **movement and the relevance of the stock definition**.

2019 (5 months) Internship in the restoration program of the seagrass *Zostera muelleri* at the Queenscliff Marine Center (Deakin University, Queenscliff, Australia), (supervision: Craig Sherman, Yi Mei Tan):

I led a project to understand the role of the microbiome in **seagrass restoration**.

I **built** a seagrass nursery to carry out **experiments** and wrote my internship report in the form of a **scientific publication**.

I helped the team for the **fieldwork** (collection of seagrass, transplantation, monitoring...), working and living with the **international** team.



List of publications:

2025

Lheureux, Edel; Buoro, Mathieu; Prévost, Etienne. "Estimating diadromous fish abundance during their migration by mark-recapture: remodeling combined with sequential Bayesian inference". In: *Canadian Journal of Fisheries and Aquatic Sciences*. DOI: [10.1139/cjfas-2024-0321](https://doi.org/10.1139/cjfas-2024-0321)

2026

Bertignac, Michel; Lheureux Edel; Drogou, Mickael; Wright, Serena; Woillez, Mathieu. "Estimating migration and mortality rates for European seabass (*Dicentrarchus labrax*) off the French coasts using electronic tagging data". In: *ICES Journal of Marine Science*. DOI: [10.1093/icesjms/fsag043](https://doi.org/10.1093/icesjms/fsag043)

In progress

Lheureux, Edel; Roy, Romain; Lauronce, Vanessa; Drouineau, Hilaire; Sagnes, Pierre. "Optimizing eel passage at large hydroelectric facilities: the contribution of multiple eel ramps."

In progress

Lheureux, Edel; Buoro, Mathieu; Prévost, Etienne. "Bayesian mechanistic approach to modelling the phenology of smolt migration: disentangling the observation process from the underlying ecological mechanisms".



List of peer review:

2025

Canadian Journal of Fisheries and Aquatic Sciences



Education:

2018-2021

Studies in fisheries science and marine ecology at Institut Agro in Rennes (French Grande Ecole, Master equivalent)

2016-2018

Preparatory class in the Lycée Pierre Corneille in Rouen to attempt to enter in a French Grande Ecole

2016

Scientific baccalauréat (High School Diploma) with honours



Skills:

- Language skills: English (TOEIC: 965 points, 5 months in Australia), Spanish (Scholar level)
- Computer skills : R, Python (beginner), Nimble, JAGS, Inkscape, LaTeX, Git , ADMB, QGIS, Artificial Intelligence Tools (chatGPT, Ellicit, Consensus)
- Driver's licence
- Coastal boating licence



Others activities & skills:

- Level 2 in scuba-diving: Diploma of the Fédération Française d'Etude et de Sport Sous-Marin (equivalence of "advanced open water diver" in padi)
- Contribution to biodiversity monitoring dives (citizen science) with the Port Vivant association (Le Havre)
- Sampling and measurements during diving for a school science project
- Contribution to the IACO coastal monitoring programme (citizen science) with the Ligue de Protection des Oiseaux (LPO)
- Sport such as cycling, taekwondo
- Drawing



International conferences:

- FSBI 2025: oral presentation
- FSBI 2023: oral presentation & convenor of a session
- Euring 2023: oral presentation
- NoWPas 2023: oral presentation & member of the organising committee
- NoWPas 2022: oral presentation